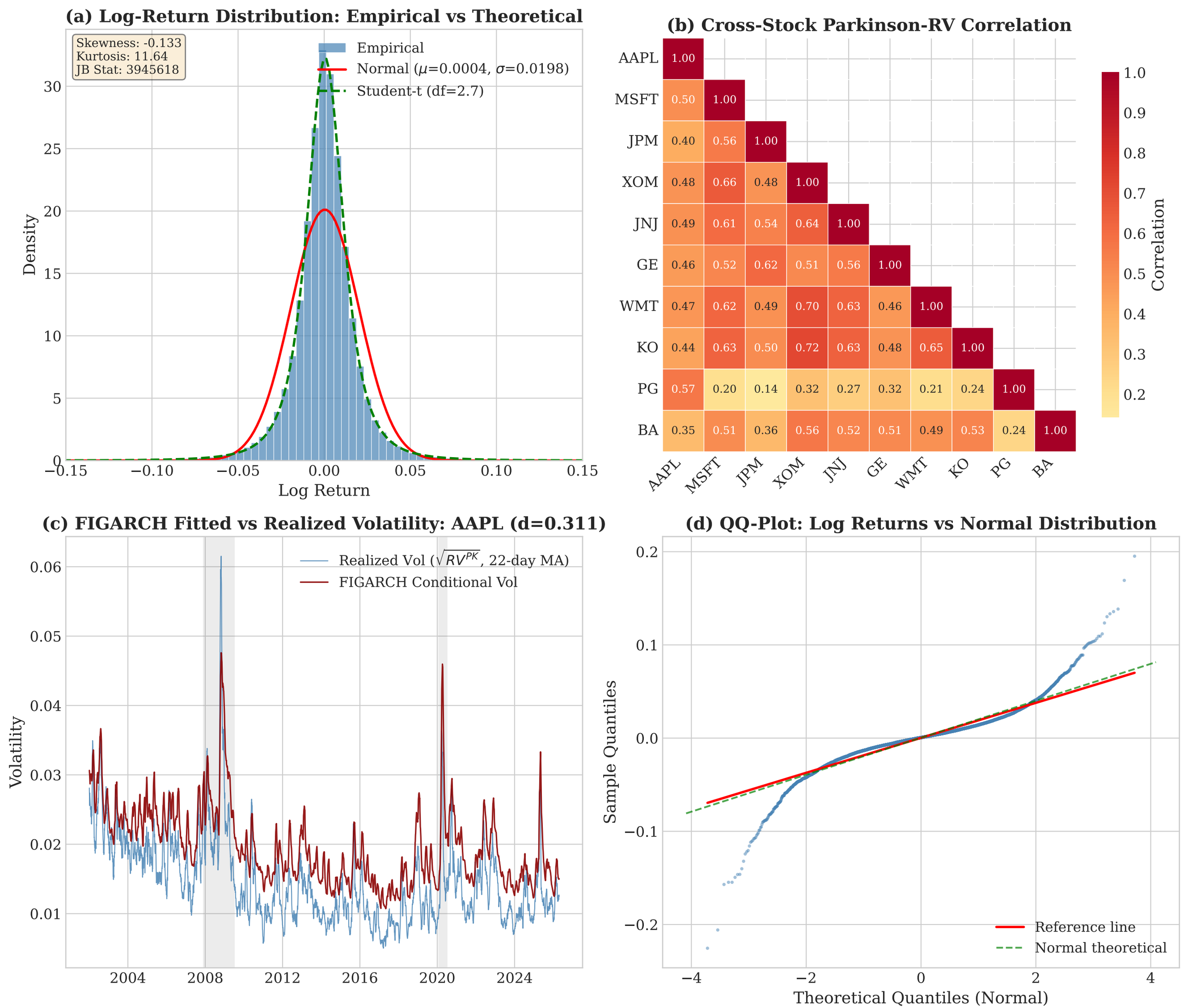
(a) Log-Return Distribution: Empirical vs Theoretical
Skewness: -0.133
Kurtosis: 11.64
JB Stat: 3945618
Empirical
Normal (μ=0.0004, σ=0.0198)
Student-t (df=2.7)
Density
Log Return
(b) Cross-Stock Parkinson-RV Correlation
AAPL 1.00
MSFT 0.50 1.00
JPM 0.40 0.56 1.00
XOM 0.48 0.66 0.48 1.00
JNJ 0.49 0.61 0.54 0.64 1.00
GE 0.46 0.52 0.62 0.51 0.56 1.00
WMT 0.47 0.62 0.49 0.70 0.63 0.46 1.00
KO 0.44 0.63 0.50 0.72 0.63 0.48 0.65 1.00
PG 0.57 0.20 0.14 0.32 0.27 0.32 0.21 0.24 1.00
BA 0.35 0.51 0.36 0.56 0.52 0.51 0.49 0.53 0.24 1.00
AAPL MSFT JPM XOM JNJ GE WMT KO PG BA
Correlation
(c) FIGARCH Fitted vs Realized Volatility: AAPL (d=0.311)
Realized Vol (√RV^PK, 22-day MA)
FIGARCH Conditional Vol
Volatility
(d) QQ-Plot: Log Returns vs Normal Distribution
Reference line
Normal theoretical
Sample Quantiles
Theoretical Quantiles (Normal)